

A finite rank window cannot show that a neural population code satisfies the eigenspectrum smoothness bound

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Abstract

Stringer, Pachitariu, Steinmetz, Carandini and Harris showed that a population code that maps a d -dimensional stimulus space differentiably to noise-free responses has an eigenspectrum decaying asymptotically faster than $n^{-(1+2/d)}$, reported exponents of 1.49, 1.65 and 3.43 for stimulus sets with $d = 8, 4$ and 1 , fitted over ranks 11–500 (ranks 5–30 for 32 grating directions in their Methods, 11–30 in their deposited code), and concluded that the code is about as high-dimensional as smoothness allows. That a finite window of ranks cannot measure an asymptotic exponent is elementary and partly anticipated; this note makes it quantitative for these stimulus sets. For codes built on bounded eigenfunctions, as on a torus, a known kernel-matrix bound implies that the finite-sample spectrum is continuous in the variance of the tail and blind to its rate of decay, so no estimator continuous in that spectrum can tell a continuously differentiable code from one with infinite expected squared gradient. We place noise-free codes of fixed smoothness (Matérn tuning) on the stimulus coordinates of all ten 8D and 4D stimulus sets. Codes exactly at the differentiability border give ranks 11–500 exponents from 0.255 to 1.628 at $d = 8$ and from 0.625 to 1.762 at $d = 4$, below the bound for short and above it for long tuning length scales, with infinitely many or 8,704 neurons and in whitened coordinates. Non-differentiable codes with Matérn $\nu = 0.75$ also exceed the bound in the unwhitened coordinates at 2,800 stimuli, but not over ranks 101–500, not in whitened 8D coordinates and not always with fewer stimuli. For 32 directions a non-differentiable code reaches 3.5012 over ranks 5–30 and 3.4306 over ranks 11–30. The tail exponent of the eigenmoment method of Pospisil and Pillow depends on the unresolved tail: spectra that share a broken power law with tail exponent 1.25 up to rank 500 give tail exponents from 1.138 to 1.394, and the eigenmoment misfit flags none of 20 simulated data sets of any of them with diagonal weights (95% upper bound 0.168 on the flag probability) and at most 5 of 20 with full whitening (upper bound 0.491), depending on the reference distribution. The reported exponents are consistent with the bound but cannot show that the code satisfies it or lies close to it.

1 Introduction

Stringer et al. [1] recorded the responses of about ten thousand neurons in mouse visual cortex to 2,800 natural images and found that the variance of the n th principal component of the stimulus-related responses decays roughly as n^{-1} . They related the decay to the smoothness of the population code: “We showed mathematically that if the sensory stimuli presented can be characterized by d parameters, and if the mapping from these parameters to (noise-free) neural population responses is differentiable, then the population eigenspectrum must decay asymptotically faster than a power law of exponent $\alpha = 1 + 2/d$ (Supplementary Discussion §2).” The stated prediction was ordinal: “We therefore hypothesized that lower-dimensional stimulus sets would evoke population responses with

larger power-law exponents.” They projected the natural images onto $d = 8$ or $d = 4$ dimensions and used drifting gratings of 32 directions for $d = 1$, and found: “Consistent with the hypothesis, stimulus sets with $d = 8, 4$, and 1 yielded power-law scaling of eigenvalues with exponents of 1.49, 1.65, and 3.43, near but above the lower bounds of 1.25, 1.50, and 3.00 predicted by the $1 + 2/d$ exponent (Fig. 3h).” The exponents come from a fixed window of ranks: “We computed the linear fit of the eigenspectrum over the range of 11 to 500 dimensions for all recordings (and model fits) other than the 32 drifting grating recordings. For the 32-grating recordings, due to noise and the length of the spectrum, we computed the power-law exponent from 5 to 30.” Their deposited code fits the gratings over ranks 11 to 30 instead (Section 3), and we report both windows. The stronger claims rest on these numbers. The Supplementary Discussion concludes: “We conclude that our experimental observations of eigenspectrum decay only just faster than $n^{-1-2/d}$ indicate a neural code that is as high-dimensional as possible before hitting the regime where these pathological conditions must occur.” The Discussion reads: “Our results therefore suggest that the eigenspectrum of the visual cortical code decays almost as slowly as possible consistent with smooth neural coding.”

That a finite window of ranks does not measure an asymptotic exponent is elementary, and parts of it are anticipated. The same Supplementary Discussion notes that for units of equal tuning radius “the eigenspectrum does not follow a power law but is approximately flat until the eigenfunctions’ spatial scale matches this radius, after which the eigenspectrum falls rapidly” (its Example 3). Pospisil and Pillow [4] showed that “cvPCA provides a biased estimator” of the spectrum, fitted parametric spectra to unbiased eigenmoment estimates (minimizing eigenmoment error, MEME), and found for the seven natural-image recordings a broken power law with “MEME average $\alpha_2 = 1.20$, cvPCA average $\alpha = 1.01$ ” for the tail. They add: “Thus it remains unclear whether the tail of the eigenspectrum decays as a power law as hypothesized by Stringer et al.” They also state that their approach “can infer eigenvalues beyond the rank of the data even if there is no relationship with early eigenvalues”, and that “We can also assess whether the parametric form is a good assumption by determining whether it can account for systematic variation in the eigenmoments”. Davidovich and Roudi [5] refitted the exponents by Bayesian interpolation and found them to depend on the noise model and the range of ranks; for the natural images the bound was not decided, while “in the case of 8D stimuli the results are consistently above the lower bound of 1.25 with a reasonable margin.” In kernel methods, the eigenvalues of a kernel matrix on P samples are known to converge to those of the integral operator as $P \rightarrow \infty$ [6], with finite-sample bounds [7], and in expectation the sum of the leading sample eigenvalues overestimates the corresponding sum of operator eigenvalues, while the sum of the trailing ones underestimates it [8].

This note makes the elementary point quantitative for the stimulus sets on which the bound was tested. It adds four things. First, a statement of the finite-stimulus eigenvalue problem and a direct consequence of the kernel-matrix bounds of [7]: for codes built on bounded eigenfunctions, the finite-sample spectrum, and each eigenmoment, is continuous in the variance of the tail and blind to its rate of decay (Section 2). Second, codes of fixed smoothness placed on the actual stimulus coordinates of all ten 8D and 4D stimulus sets, with the dependence of the window exponent on the tuning length scale, the rank window, the metric, the number of stimuli and the number of neurons. Third, for 32 grating directions, the split of the window shortfall into a pre-asymptotic part and an aliasing part. Fourth, the sensitivity of the MEME tail exponent to changes of the spectrum beyond the resolved ranks, and what the eigenmoment misfit can and cannot detect. Proposition 1 and Corollary 1 are proved; every other result is numerical, computed in double precision except for the single-precision steps of the simulations stated in Section 3, and none is stated as a theorem. We fitted no exponent to any recording. The only neural data used are the per-neuron signal and noise variances of one natural-image recording, which calibrate the simulator of Section 3, and that recording’s cvPCA spectrum at ranks 11, 100 and 500, with which Section 4.6 compares simulated

spectra. We make no claim that the reported exponents are wrong or that the codes are not smooth; the question is only what exponents fitted over these windows can show.

2 The finite-stimulus eigenvalue problem

Codes and spectra. Let stimuli s be drawn from a distribution μ on a stimulus space S , and let a population code map each stimulus to a noise-free response vector $\varphi(s)$ in a real Hilbert space $\mathcal{H}$ (for N neurons, $\mathcal{H} = \mathbb{R}^N$). Write $\bar{\varphi} = \mathbb{E}_s \varphi(s)$ and $k(s, s') = \langle \varphi(s) - \bar{\varphi}, \varphi(s') - \bar{\varphi} \rangle$. The population eigenspectrum $\lambda_1 \geq \lambda_2 \geq \dots$ is the spectrum of the covariance operator of $\varphi(s)$, and its nonzero part equals the spectrum of the integral operator $(Tf)(s) = \int k(s, s') f(s') d\mu(s')$ on $L^2(\mu)$. Theorem 5 of the Supplementary Discussion of [1] takes S to be a closed compact d -dimensional Riemannian manifold with its canonical measure and states that if the expected squared derivative $\mathbb{E}_s \|\nabla_s \varphi(s)\|^2$ is finite, then $\lambda_n = o(n^{-1-2/d})$. Its proof writes $\mathbb{E}_s \|\nabla_s \varphi\|^2 = \sum_i \lambda_i \mathbb{E}_s |\nabla v_i|^2$ for the kernel eigenfunctions v_i and uses Weyl’s law for the Laplace–Beltrami eigenvalues, which grow as $n^{2/d}$. Theorem 5 is stated for the uncentred kernel $\langle \varphi(s), \varphi(s') \rangle$, whose integral operator has the nonzero spectrum of the second-moment operator $\mathbb{E}_s \varphi(s) \otimes \varphi(s)$ on $\mathcal{H}$. That operator exceeds the covariance operator by the positive rank-one operator $\bar{\varphi} \otimes \bar{\varphi}$, so by the minimax principle the $(n+1)$ th eigenvalue of either is at most the n th of the other, and the conclusion $o(n^{-1-2/d})$ holds for both.

Finitely many stimuli. An experiment presents P stimuli $s_1, \dots, s_P$. In the noise-free limit the sample covariance of the population responses, with each neuron centred over the stimuli, has the same nonzero eigenvalues as the $P \times P$ matrix $P^{-1} H K H$, where $K_{ab} = \langle \varphi(s_a), \varphi(s_b) \rangle$ and $H = I - P^{-1} \mathbf{1} \mathbf{1}^\top$. The finite-stimulus eigenvalue problem is

$$\frac{1}{P} \sum_{b=1}^P (H K H)_{ab} v_b = m v_a, \quad a = 1, \dots, P, \quad (1)$$

with eigenvalues $m_1 \geq \dots \geq m_P = 0$, the Nyström discretization of T at the sample points. The window exponent is the slope of $\log m_n$ against $-\log n$ over a window of ranks, fitted as in [1] by least squares with weights $1/n$; for the natural-image and low-dimensional sets the window is $11 \leq n \leq 500$, and for 32 gratings ranks 5–30 or 11–30 (Section 3). Estimators such as cross-validated PCA (cvPCA) target m_n from noisy data and return it in the noise-free limit, so what follows applies to any estimator of m_n .

Proposition 1. *Let $\psi_1, \psi_2, \dots$ be functions on S with $\sup_s |\psi_j(s)| \leq C$ for all j , and let (λ_j) and (λ'_j) be nonnegative summable sequences with $\lambda_j = \lambda'_j$ for $j \leq M$. Define $k(s, s') = \sum_j \lambda_j \psi_j(s) \psi_j(s')$ and k' likewise, and write $\tau_M = \sum_{j>M} \lambda_j$, $\tau'_M = \sum_{j>M} \lambda'_j$. Then for every P and every choice of $s_1, \dots, s_P$ the eigenvalues of (1) for k and k' satisfy $|m_n - m'_n| \leq C^2 \max(\tau_M, \tau'_M)$ for all n .*

Proof. The series converge absolutely and uniformly because $|\psi_j| \leq C$. The matrices of (1) for k and k' differ by $P^{-1} H (B - B') H$, where $B_{ab} = \sum_{j>M} \lambda_j \psi_j(s_a) \psi_j(s_b)$ and B' likewise. Both B and B' are positive semidefinite, so $\|B - B'\| \leq \max(\|B\|, \|B'\|)$, and $\|B\| \leq \text{tr } B = \sum_{j>M} \lambda_j \sum_a \psi_j(s_a)^2 \leq P C^2 \tau_M$. Multiplying by the projection H on both sides does not increase the operator norm, so the difference has norm at most $C^2 \max(\tau_M, \tau'_M)$. By Weyl’s inequality (the Courant–Fischer minimax characterization), the n th eigenvalues of two symmetric matrices differ by at most the operator norm of their difference. $\square$

The proof adapts the argument of Lemmas 5 and 8 of Braun [7]. Braun states them for the uncentred kernel matrix of a Mercer kernel whose eigenfunctions are orthonormal in $L^2(\mu)$ and whose eigenvalues are nonincreasing (his Section 2.1): Lemma 5 is Weyl's inequality for a kernel matrix and that of its truncation, and Lemma 8 bounds the norm of the truncation error by M^2 times the tail sum of the eigenvalues, M being a uniform bound on the eigenfunctions. In that setting, applied to each of two kernels with a common head and combined by the triangle inequality, they give the bound for the uncentred matrices with $\tau_M + \tau'_M$ in place of $\max(\tau_M, \tau'_M)$. The proof above applies Weyl's inequality to the centred matrices directly, uses that H does not increase the norm, and uses the positivity of B and B' for the maximum; it needs neither orthonormality, nor zero mean, nor any ordering of the ψ_j and λ_j . If the ψ_j are orthonormal with mean zero in $L^2(\mu)$ and the λ_j are nonincreasing, then (λ_j) is the population spectrum of a code with kernel k . The head of the spectrum fixes the finite-sample eigenvalues up to $C^2\tau_M$, whereas the smoothness bound concerns only how the tail decays, which τ_M does not determine. Two codes can trivially be made to agree at a given set of stimuli by changing one of them elsewhere; the proposition is stronger in that it holds uniformly, for every stimulus set of every size.

Corollary 1. *Let S be the torus $\mathbb{R}^d/\mathbb{Z}^d$ with the uniform measure and ψ_j the nonconstant real Fourier functions $\sqrt{2}\cos(2\pi\kappa \cdot s)$ and $\sqrt{2}\sin(2\pi\kappa \cdot s)$, with $\kappa \in \mathbb{Z}^d \setminus \{0\}$ taken once from each pair $\pm\kappa$ and ordered by $|\kappa|$, so that $C = \sqrt{2}$ and ψ_j is a Laplacian eigenfunction with eigenvalue $\eta_j = 4\pi^2|\kappa_j|^2$. For any P , any stimuli, any M and any $\delta > 0$ there are two codes whose spectra agree for $j \leq M$, one continuously differentiable and one with $\mathbb{E}_s\|\nabla\varphi\|^2 = \infty$, whose finite-sample eigenvalues (1) differ by less than δ at every rank.*

Proof. Take $\varphi(s) = \sum_j \sqrt{\lambda_j} \psi_j(s) e_j$ for an orthonormal family e_j ; its kernel is k and $\mathbb{E}_s\|\nabla\varphi\|^2 = \sum_j \lambda_j \eta_j$. The number of lattice points with $|\kappa| \leq R$ grows as $\omega_d R^d$, with ω_d the volume of the unit ball, so $\eta_j \sim 4\pi^2(j/\omega_d)^{2/d}$. Take $\lambda_j = j^{-b}$ for $j \leq L$ with some $b > 1$, and for $j > L$ continue with $\lambda_j = L^{-b}(j/L)^{-a}$ in one code and $L^{-b}(j/L)^{-a'}$ in the other, with $1 < a < 1 + 2/d < a'$. The first code has $\sum_j \lambda_j \eta_j = \infty$. For the second the sum is finite, and since $|\nabla\psi_j|^2 \leq 2\eta_j$ the series of derivatives converges uniformly, so this code is continuously differentiable as a map into $\mathcal{H}$. Both tail masses beyond L are at most $\int_L^\infty L^{-b}(x/L)^{-a} dx = L^{1-b}/(a-1)$, which is below $\delta/2$ for $L \geq M$ large enough, and Proposition 1 with M replaced by L and $C^2 = 2$ gives the claim. $\square$

For the code of Corollary 1 with infinite expected squared gradient, which need not be stationary, we claim no more than that. For a stationary code, whose kernel $\langle \varphi(s), \varphi(s') \rangle = k(s - s')$ has spectral measure F , an infinite expected squared gradient means that the code is differentiable nowhere. For a unit vector e , $\|\varphi(s + te) - \varphi(s)\|^2 = 2(k(0) - k(te)) = \int 2(1 - \cos(t\omega \cdot e)) dF(\omega)$, and Fatou's lemma gives $\liminf_{t \rightarrow 0} \|\varphi(s + te) - \varphi(s)\|^2/t^2 \geq \int (\omega \cdot e)^2 dF(\omega)$, which for $d = 1$ or an isotropic F equals $d^{-1} \int |\omega|^2 dF(\omega) = d^{-1} \mathbb{E}_s\|\nabla\varphi\|^2$. When this is infinite the difference quotients are unbounded at every s , so φ has no derivative anywhere as a map into $\mathcal{H}$. This covers the Matérn codes below with $\nu \leq 1$ and the circle codes of Section 3 whose spectrum satisfies $\sum_k k^2 c_k = \infty$; we call these codes non-differentiable.

For codes on bounded eigenfunctions, then, no estimator that depends continuously on the finite-sample spectrum, and none that is robust to arbitrarily small changes of it, can decide the bound without an assumption about the tail, even before any estimation error (Section 4.4 gives numerical instances on the circle). The same holds for the eigenmoments $\text{tr } \Sigma^p = \sum_j \lambda_j^p$ that MEME fits: for nonincreasing spectra that agree for $j \leq M$ they differ by at most $\sum_{j>M} (\lambda_j^p + \lambda'_j{}^p) \leq \lambda_{M+1}^{p-1} \tau_M + \lambda'_{M+1}{}^{p-1} \tau'_M$, so for $p \geq 2$ the tail variance enters weighted by the $(p-1)$ th power of eigenvalues beyond the head, and the trace ($p = 1$) records the tail variance but not its decay.

This suggests, but does not show, that a tail exponent fitted to the moments mostly continues the assumed parametric form: the inequality bounds how much the moments can change, not what a fit to estimated moments returns. Section 4.5 tests this numerically. We do not prove the analogue for the Matérn codes on $\mathbb{R}^d$ below, whose eigenfunctions need not be bounded; for them we compute.

Matérn codes. To see what happens for smooth, stationary codes, consider a population whose tuning curves are independent draws of a Gaussian process with the Matérn covariance

$$k_{\nu,\ell}(r) = \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu} r}{\ell} \right)^\nu K_\nu \left(\frac{\sqrt{2\nu} r}{\ell} \right), \quad r = \|s - s'\|, \quad (2)$$

where K_ν is the modified Bessel function of the second kind; with infinitely many neurons the kernel of the code is $k_{\nu,\ell}$ up to a constant factor. Its spectral density is proportional to $(2\nu/\ell^2 + |\omega|^2)^{-(\nu+d/2)}$, so $\mathbb{E}_s \|\nabla \varphi\|^2 = -\Delta k_{\nu,\ell}(0)$ is proportional to $\int |\omega|^2 (2\nu/\ell^2 + |\omega|^2)^{-(\nu+d/2)} d\omega$, which is finite if and only if $\nu > 1$, whatever the stimulus distribution. We write $\alpha_\infty = 1 + 2\nu/d$. On the torus the lattice count above gives $\lambda_n \sim n^{-\alpha_\infty}$; for a stimulus density p on $\mathbb{R}^d$ the Weyl-type phase-space count gives the same exponent if $\int p^{d/(2\nu+d)} < \infty$, which we assume and do not prove, and we use α_∞ only to label codes. The code with $\nu = 1$ sits at the border: $\alpha_\infty = 1 + 2/d$, and its expected squared gradient diverges logarithmically, so it fails the finite-gradient condition of Theorem 5 (whose compact-manifold hypothesis these stimulus sets do not meet in any case), while every $\nu > 1$ satisfies it. The length scale ℓ does not enter the bound.

3 Methods

Stimulus coordinates. The deposit of Stringer et al. [2] contains, for each of the six 8D and four 4D recordings, the 2,800 stimulus images shown (68×270 pixels, 8-bit integers). The stimulus coordinates use these image files and no neural data. For each set we centred the pixel matrix and took the top d principal components as coordinates, without whitening. They carry at least 0.9996 of the pixel variance, and the next component has at most 7.2×10^{-6} of the variance of the first, consistent with rounding to 8 bits. Table 1 lists, for each set, the relative variances of the d components, their participation ratio $(\sum_i v_i)^2 / \sum_i v_i^2$, the median nearest-neighbour distance relative to the median pairwise distance, and the standard deviation of the smallest coordinate relative to the median nearest-neighbour distance. As a property of the stimuli alone we also computed their nearest-neighbour dimension d_{nn} , defined as in [10] by the scaling of the median nearest-neighbour distance of n random stimuli as $n^{-1/d_{\text{nn}}}$ (least squares on logarithms, $n = 175, 350, 700, 1,400$ and $2,800$, 4 subsets per n), and, as a control for finite n , the same for 2,800 Gaussian points with the set’s coordinate variances.

Kernel spectra. For each set, $\nu \in \{0.5, 0.75, 1, 1.5, 2.5\}$ and $\ell \in \{1/8, 1/4, 1/2, 1, 2, 4, 8\}$ times the median pairwise distance, we formed the $2,800 \times 2,800$ matrix of (2) and computed all eigenvalues of (1) in double precision. The window exponent is the weighted least-squares slope of [1] (`powerlaws/get_powerlaw.m` of [3], weights $1/n$ at the 1-based ranks n) over ranks 11–500, and also over ranks 11–100 and 101–500. In every spectrum m_{500}/m_1 exceeds 1.6×10^{-9} , far above rounding error. These computations are deterministic, with no sampling error. Two further versions test the choices behind them. In whitened coordinates each of the d coordinates is scaled to unit variance, so that all stimulus parameters count equally ($\nu = 0.75$ and 1 at every ℓ , and $\nu = 0.5, 1.5, 2.5$ at $\ell = 1/4$ and 4). For the dependence on the number of stimuli we took 4 random subsets of $P = 1,000, 1,400$ and $2,000$ stimuli of each set, keeping the code fixed ($\nu = 0.75$ and 1 at $\ell = 1, 4, 8$); the subsets sample the same kernel operator, so the change with P is a finite-sample effect.

Finite populations. With N neurons whose tuning curves are independent Gaussian-process draws, the centred response vectors Hf_i are independent $\mathcal{N}(0, HKH)$, so the population spectrum is distributed exactly as the eigenvalues of $D^{1/2}WD^{1/2}$, where D holds the eigenvalues of (1) and W is a Wishart matrix with identity scale and N degrees of freedom, divided by N . We drew W by the Bartlett decomposition with $N = 8,704$ (the neuron count of the recording that calibrates the simulator below), the same draw for every kernel within a replicate, for $\nu = 1$ at every $\ell \geq 1/4$ and $\nu = 0.75$ at $\ell = 2, 4, 8$ on all ten sets, and for all five ν at $\ell = 1/4, 1, 4$ on two sets (8D MP032 2017-08-10, 4D MP032 2017-09-22). Cells of the first computation have 20 replicates and cells added in revision 5; the between-population standard deviation is 0.0013 to 0.0021, so either count fixes the mean well.

Gratings, $d = 1$. A stationary code on the circle of directions has Fourier eigenfunctions; we took the kernel spectrum $c_k = (1 + k^2/\kappa^2)^{-(\nu+1/2)}$, $k \in \mathbb{Z}$, the circle’s counterpart of (2), with bandwidth κ (tuning width of order $2\pi/\kappa$), $\alpha_\infty = 1 + 2\nu$ and operator eigenvalues $c_1, c_1, c_2, c_2, \dots$. For $P = 32$ equally spaced directions the centred matrix K/P is circulant, with eigenvalues $\sum_{k \equiv j \pmod{32}} c_k$, $j = 1, \dots, 31$ (sums over $|k| \leq 2^{22}$ plus the integral of the remainder). The Methods of [1] fit ranks 5–30, but their deposited code at the cited commit fits ranks 11–30: `mainfigs/fig3.m`, line 91, sets `trange0 = [11:30]` for the gratings (line 94 sets `11:500` for the other stimulus sets), and `powerlaws/statsShuffledPCA.m`, line 59, fits ranks 11 to $\min(500, n - 2)$ of an n -value spectrum, which is 11–30 for 32 gratings. The code fits the reported exponent to the mean of the normalized spectra over the grating recordings (`fig3.m`, line 111), and the published 3.43 was most probably fitted over ranks 11–30. We fitted both windows, ranks 5–30 and ranks 11–30, of the operator spectrum and of the 32-direction spectrum for $\nu \in \{0.5, 1, 1.5, 2\}$ and $\kappa \in \{0.5, 1, 2, 4, 8, 16\}$, and repeated the finite-population computation with $N = 8,704$ in 2,000 replicates.

Eigenmoment fits. MEME [4] estimates the signal eigenmoments $\text{tr } \Sigma^p$ without bias from cross-repeat products of differences of stimulus pairs, extending the estimator of Kong and Valiant [9] to noisy data, and fits a parametric spectrum over as many indices as there are neurons; the broken power law has exponent α_1 before a break and α_2 after it. We used our own implementation (`est.py` in the source folder, written for an earlier, unpublished feasibility study in this project whose results are not used or reported here), which follows the authors’ code [11] with three deviations: eight moments, least squares on log moments, and diagonal weights (1/SD of each log moment) instead of a whitening matrix. The break is profiled over $\{4, 6, 8, 10, 12, 15, 20, 30\}$ and the fit runs over $N = 8,704$ indices. The base spectrum is a broken power law with $\lambda_1 = 200$, exponent 0.5 up to rank 10 and $\alpha_2 = 1.25$ or 1.5 after it. Five variants agree with it up to rank 500 and continue with exponent 0.8, 1.0, 2.0 or 3.0, or drop to zero after rank 2,800; three more change every rank slightly (a constant floor adding 5% or 15% of the base variance, which is 4.8% or 13% of the total, or a factor $e^{-n/1000}$). We fitted their exact moments $\sum_n \lambda_n^p$, $p = 1, \dots, 8$, and moments estimated from simulated data. The simulated data follow the noise model of the simulator written for the same study (`sim.py`): 2,800 stimuli, two repeats, 8,704 neurons, noise equal to 0.75 times the signal covariance plus one shared mode of variance 590 plus independent per-neuron noise whose variances are set from the per-neuron noise and signal variances of the natural-image recording M170714_MP032 2017-09-14. Neuron gains are equal here, so the signal spectrum is exactly the one specified. We simulated 20 data sets for the base ($\alpha_2 = 1.25$) and its five far-tail variants, with the same random numbers for every spectrum within a data set, and 80 further data sets of the base alone. The weights are 1/SD of the log moment estimates over those 80 data sets, independent of the 20 paired ones. The data sets redraw stimuli and noise but keep the neurons fixed; this understates

the moment variances and so favours detecting a change. The misfit of a fit is its sum of squared weighted residuals. Its null distribution is that of the correctly specified model over all 100 base data sets, each fitted with weights estimated from the other four fifths; a fit is flagged when its misfit exceeds the 95th percentile of this distribution. As a check closer to Pospisil and Pillow’s method we repeated everything with full-covariance whitening of the log moments (covariance estimated from the same data sets; condition number 7.7×10^7). For the Matérn codes we computed the eigenmoment estimates that MEME would obtain from noise-free responses at the 2,800 stimuli with infinitely many neurons, by applying its estimator, which without noise is the estimator of [9] applied to differences of stimulus pairs, to the kernel matrix itself (for $\nu \in \{0.75, 1, 1.5\}$ and $\ell \in \{1/4, 1, 4\}$); these estimate the population moments over the stimulus distribution, which is MEME’s target, unlike the moments of the 2,800-stimulus sample spectrum. At $\ell = 1/4$ we repeated this for a finite population of 8,704 Gaussian-process neurons, whose Gram matrix is $LWL^\top$ with $LL^\top = K$ and W a Wishart matrix as above (one draw per stimulus set). These fits use the weights and the misfit threshold of the simulated broken power law.

cvPCA and noise level. As a side result we computed the cvPCA window bias against the noise level with the same simulator: the broken power law $\alpha_2 = 1.5$ and heterogeneous gains (realized window slope 1.497), aligned noise unchanged and the other noise variance divided by $k \in \{1, 4, 16\}$, 8 replicates per level. For comparison we use the cvPCA spectrum of the calibration recording at ranks 11, 100 and 500, which the stage-1 program `calib.py` computes from its two repeats; no exponent is fitted to it.

Precision. Spectra, fits and window exponents are computed in double precision. The simulator draws responses in single precision, and the cross-repeat Gram products of the MEME and cvPCA simulations and the eigendecomposition inside cvPCA are formed in single precision. Redoing 3 base data sets of the MEME simulation and two cvPCA replicates with the same random draws in double precision throughout changed the log eigenmoments by at most 1.2×10^{-6} , the squared whitened residual shift by at most 4.1×10^{-10} , α_2 by at most 1.2×10^{-7} , the misfit by at most 5.0×10^{-5} and the cvPCA window exponent by at most 3.1×10^{-7} , all far below the printed precision (`precision_check.py`).

Uncertainty. Deterministic results carry no error bars. Simulated ones are given as mean $\pm$ standard error, with the standard deviation over data sets where a single data set matters, and paired differences with 95% t intervals. Proportions of data sets (flag rates) are given as counts out of 20 with exact (Clopper–Pearson) 95% intervals.

4 Results

4.1 Stimulus geometry

The four 4D sets are alike (Table 1), with participation ratios 3.1 to 3.6. The six 8D sets are not: their participation ratios range from 3.9 to 6.8, and in two of them (MP032 2017-08-10, MP033 2017-08-22) the last two or three coordinates carry less than 2% of the variance of the first. In every 8D set the smallest coordinate has a standard deviation of only 0.11 to 0.64 times the median nearest-neighbour distance, itself 0.14 to 0.27 of the median pairwise distance: 2,800 stimuli do not sample short distances in eight dimensions, and smoothness is a short-distance property. At the scales these stimuli resolve, the sets are also lower-dimensional than nominal: the nearest-neighbour

Table 1: Geometry of the ten stimulus sets, from the image files. Relative variances of coordinates 2, ..., d (the first is 1); PR, participation ratio of the d variances; NN, median nearest-neighbour distance over the median pairwise distance; last, standard deviation of the smallest coordinate over the median nearest-neighbour distance.

set	mouse, date	relative variances	PR	NN	last
8D	MP030 06-07	0.912, 0.733, 0.627, 0.496, 0.340, 0.328, 0.251	6.7	0.260	0.64
8D	MP031 07-02	0.793, 0.675, 0.545, 0.345, 0.286, 0.253, 0.047	5.9	0.238	0.33
8D	MP032 08-10	0.572, 0.549, 0.350, 0.261, 0.180, 0.012, 0.002	4.6	0.180	0.11
8D	MP032 09-15	0.793, 0.717, 0.573, 0.443, 0.310, 0.266, 0.107	6.2	0.245	0.47
8D	MP033 08-22	0.835, 0.497, 0.412, 0.089, 0.019, 0.008, 0.005	3.9	0.136	0.23
8D	MP034 09-15	0.807, 0.726, 0.674, 0.549, 0.440, 0.319, 0.223	6.8	0.266	0.58
4D	MP032 09-22	0.711, 0.612, 0.315	3.5	0.071	3.46
4D	MP033 09-19	0.925, 0.570, 0.111	3.1	0.059	2.50
4D	MP033 09-22	0.699, 0.608, 0.391	3.6	0.071	3.83
4D	MP034 09-20	0.881, 0.568, 0.284	3.4	0.063	3.61

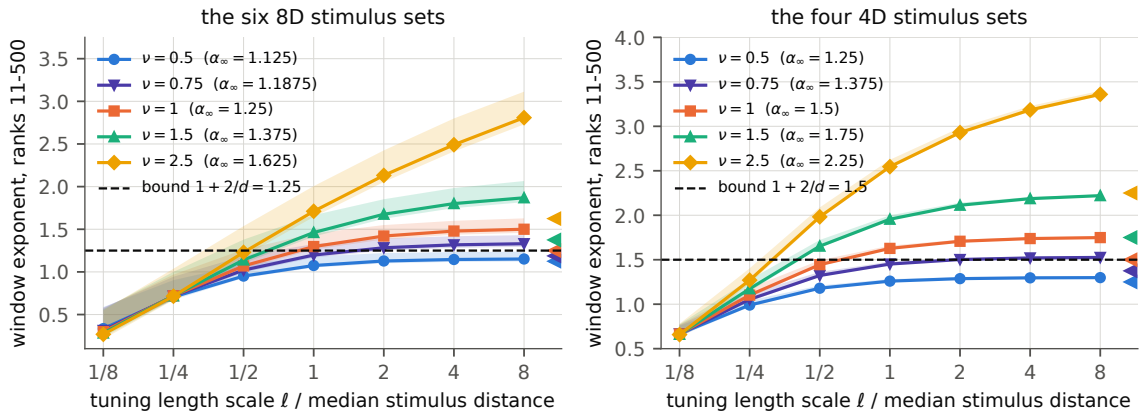


Figure 1: Ranks 11–500 window exponent of noise-free Matérn codes (infinitely many neurons) on the real stimulus coordinates, against the tuning length scale. Lines: median over the stimulus sets; bands: range over the sets. Arrowheads on the right: $\alpha_\infty = 1 + 2\nu/d$. Dashed line: the bound $1 + 2/d$. Deterministic. Table 3 gives the effect of finite populations, the rank window, the metric and the number of stimuli.

dimension is 2.8 to 3.0 in the 4D sets and 5.2 to 7.0 in the 8D sets (2.9 to 3.0 and 6.5 to 7.4 in whitened coordinates), against 3.8 to 4.3 and 6.1 to 7.9 for Gaussian points with the same coordinate variances and the same numbers of points (3.9 to 4.2 and 7.2 to 8.1 for isotropic ones, the control for whitened coordinates).

4.2 Window exponents of Matérn codes

With infinitely many neurons the border codes ($\nu = 1$) give window exponents from 0.255 to 1.628 at $d = 8$ (bound 1.25) and from 0.625 to 1.762 at $d = 4$ (bound 1.50) (Figure 1, Table 2). They lie below the bound in every stimulus set for $\ell \leq 1/4$ at $d = 8$ and $\ell \leq 1/2$ at $d = 4$, and above it in every set for $\ell \geq 1$ at $d = 8$ and $\ell \geq 1$ at $d = 4$. The smoothness of these codes is fixed; the length scale moves the exponent across the bound, as Example 3 of [1] anticipates for a single tuning radius. The stimulus set matters as well: at $\ell = 1$ the border code gives 1.261 to 1.451 across the 8D sets, the larger values in the sets with the smaller participation ratios, and 1.622 to 1.659 across the 4D sets. Differentiable codes fall below the bound at short length scales: at $\ell = 1/4$ the largest

Table 2: Ranks 11–500 window exponent of noise-free Matérn codes (infinitely many neurons) on the real stimulus coordinates: range over the six 8D and the four 4D stimulus sets. Deterministic; no sampling error.

d	ν	α_∞	length scale ℓ / median pairwise distance						
			1/8	1/4	1/2	1	2	4	8
8	0.5	1.125	0.30–0.59	0.67–0.90	0.93–1.09	1.05–1.18	1.10–1.22	1.12–1.23	1.12–1.24
8	0.75	1.1875	0.27–0.58	0.68–0.94	0.99–1.19	1.17–1.33	1.25–1.39	1.28–1.42	1.29–1.43
8	1	1.25	0.26–0.57	0.68–0.97	1.04–1.26	1.26–1.45	1.38–1.55	1.44–1.60	1.46–1.63
8	1.5	1.375	0.23–0.56	0.67–1.00	1.10–1.38	1.41–1.67	1.62–1.85	1.75–1.98	1.82–2.07
8	2.5	1.625	0.21–0.55	0.67–1.04	1.18–1.54	1.64–2.01	2.05–2.42	2.41–2.80	2.73–3.11
4	0.5	1.25	0.64–0.76	0.97–1.04	1.17–1.21	1.25–1.28	1.28–1.30	1.29–1.31	1.30–1.31
4	0.75	1.375	0.63–0.77	1.03–1.12	1.31–1.36	1.45–1.48	1.50–1.52	1.52–1.53	1.52–1.54
4	1	1.5	0.62–0.77	1.08–1.18	1.43–1.49	1.62–1.66	1.70–1.73	1.73–1.75	1.75–1.76
4	1.5	1.75	0.62–0.77	1.14–1.28	1.63–1.72	1.95–2.00	2.11–2.14	2.18–2.21	2.21–2.23
4	2.5	2.25	0.61–0.78	1.23–1.40	1.96–2.09	2.53–2.62	2.91–2.98	3.17–3.23	3.34–3.40

Table 3: Ranks 11–500 window exponent of the border codes ($\nu = 1$) and of codes below the border ($\nu = 0.75$) under changes of the computation: range over the stimulus sets, with the number of sets above the bound $1 + 2/d$ in parentheses. $N = \infty$: Table 2. $N = 8,704$: mean over finite populations. 101–500: the same spectra fitted over ranks 101–500. Whitened: coordinates scaled to unit variance. P : mean over 4 random subsets of the stimuli.

d	ν	ℓ	$N = \infty$	$N = 8,704$	ranks 101–500	whitened	$P = 1,400$	$P = 2,000$
8	1	1	1.26–1.45 (6)	1.24–1.44 (5)	1.19–1.35 (2)	1.13–1.19 (0)	1.20–1.41 (2)	1.24–1.43 (4)
8	1	4	1.44–1.60 (6)	1.42–1.59 (6)	1.30–1.43 (6)	1.30–1.36 (6)	1.38–1.56 (6)	1.41–1.58 (6)
8	1	8	1.46–1.63 (6)	1.44–1.62 (6)	1.31–1.44 (6)	1.33–1.38 (6)	1.41–1.59 (6)	1.44–1.61 (6)
8	0.75	4	1.28–1.42 (6)	1.26–1.40 (6)	1.13–1.24 (0)	1.17–1.21 (0)	1.21–1.36 (3)	1.25–1.39 (6)
8	0.75	8	1.29–1.43 (6)	1.27–1.41 (6)	1.13–1.25 (0)	1.18–1.23 (0)	1.22–1.37 (4)	1.26–1.40 (6)
4	1	1	1.62–1.66 (4)	1.62–1.66 (4)	1.62–1.64 (4)	1.60–1.63 (4)	1.60–1.65 (4)	1.61–1.65 (4)
4	1	4	1.73–1.75 (4)	1.73–1.75 (4)	1.66–1.67 (4)	1.74–1.75 (4)	1.72–1.74 (4)	1.73–1.75 (4)
4	1	8	1.75–1.76 (4)	1.74–1.76 (4)	1.66–1.67 (4)	1.76–1.76 (4)	1.73–1.75 (4)	1.74–1.76 (4)
4	0.75	4	1.52–1.53 (4)	1.51–1.53 (4)	1.43–1.44 (0)	1.53–1.53 (4)	1.48–1.50 (1)	1.50–1.52 (4)
4	0.75	8	1.52–1.54 (4)	1.51–1.53 (4)	1.43–1.44 (0)	1.53–1.54 (4)	1.49–1.51 (1)	1.51–1.52 (4)

exponent over the 8D sets is 1.001 for $\nu = 1.5$ and 1.037 for $\nu = 2.5$, and over the 4D sets 1.276 and 1.403. At long length scales they overshoot their own asymptotic exponent ($\nu = 2.5$ at $d = 4$ reaches 3.399 against $\alpha_\infty = 2.25$). Codes of fixed smoothness also reproduce the ordinal pattern that Stringer et al. predicted: in all 35 combinations of ν and ℓ , every 4D set gives a larger exponent than every 8D set.

4.3 How far these statements carry

Table 3 tests these statements against the choices behind them. With 8,704 neurons the window exponents are mostly lower: over the 112 computed cells the mean shift ranges from -0.050 to $+0.007$, and the 8 cells that rise are codes with $\nu = 1.5$ or 2.5 ; between populations the standard deviation is only 0.0013 to 0.0021. The border codes then lie below the bound in every set for $\ell \leq 1/2$ at $d = 8$ and $\ell \leq 1/2$ at $d = 4$, and above it in every set for $\ell \geq 2$ at $d = 8$ and $\ell \geq 1$ at $d = 4$ ($\ell = 1/8$ was not computed at finite N); at $\ell = 1$ they give 1.2424 to 1.4394 across the 8D sets, with MP030 2017-06-07 (1.2502; 2.5–97.5% over populations 1.2479–1.2524) and MP034 2017-09-15 (1.2424; 2.5–97.5% over populations 1.2400–1.2446) within 0.005 of the bound or below

it. In the unwhitened coordinates they reach the reported exponents, 1.49 at $d = 8$ and 1.65 at $d = 4$, in three of the six 8D sets and in all four 4D sets, over the length scales computed (up to $\ell = 8$). The 8D count rests on a small margin: MP031 2017-07-02 reaches 0.0035 above 1.49, with a standard deviation of 0.0021 over its 5 populations. In whitened coordinates no 8D border code reaches 1.49: the largest value, with infinitely many neurons, is 1.385, at $\ell = 8$, the longest length scale computed. The 4D sets still reach 1.65 there (1.759 to 1.760 at $\ell = 8$). For non-identification an existence statement in one metric suffices, but the 8D count holds only in the unwhitened metric.

For $\nu = 0.75$ the expected squared gradient diverges and $\alpha_\infty = 1 + 1.5/d$ lies below the bound (1.1875 at $d = 8$, 1.375 at $d = 4$). Over ranks 11–500, in the unwhitened coordinates and at 2,800 stimuli, these codes nevertheless exceed the bound at $\ell = 4$ in every set, with infinitely many neurons (1.282 to 1.419 at $d = 8$, 1.517 to 1.533 at $d = 4$) and with 8,704 neurons in all six 8D sets and all four 4D sets. This is an existence statement for that window, that metric and that number of stimuli, and each matters. Over ranks 101–500 all 29 cells of $\nu = 0.75$ that lie above the bound over ranks 11–500 fall below it (at $\ell = 4$: 1.125 to 1.241 at $d = 8$, 1.433 to 1.443 at $d = 4$). The spectra are curved: for $\ell \geq 1$ at $d = 8$ the ranks 11–100 exponent exceeds the ranks 101–500 exponent by a median of 0.21 (over all cells the difference ranges from -0.76 to $+0.83$), and for the 7 border-code spectra that reach 1.49 in the 8D sets by 0.24 to 0.33. We did not compute sub-window exponents of the recorded spectra, so whether spectra of this shape could stand in for the recorded ones is not tested here; the argument needs only that they give the same ranks 11–500 exponent. In whitened coordinates the $\nu = 0.75$ codes at $\ell = 4$ give 1.168 to 1.214 at $d = 8$, above the bound in none of the six sets, while the 4D sets hardly change (1.525 to 1.531). The border codes keep landing on both sides in whitened coordinates: at $d = 8$ they give 1.130 to 1.185 at $\ell = 1$ and 1.332 to 1.385 at $\ell = 8$. With fewer stimuli the exponents fall: at $\ell = 4$ the 8D sets give 1.212 to 1.358 at $P = 1,400$ (above the bound in three of six sets) and 1.252 to 1.391 at $P = 2,000$ (above the bound in six; in MP034 2017-09-15 by only 0.0024, with a standard deviation over subsets of 0.0020 and one subset at 1.2498). The subsets sample the same operator, so at 2,800 stimuli the window exponent still depends on the number of stimuli; its limit as P grows, the window exponent of the operator eigenvalues at ranks 11–500, cannot be computed from these stimulus sets. Solving $w = 1 + 2\nu/d_{\text{eff}}$ for d_{eff} at $\ell = 8$ gives 3.2 to 4.4 in the 8D sets and 2.6 to 2.7 in the 4D sets for the border codes, and 3.5 to 5.1 and 2.8 to 2.9 for $\nu = 0.75$. Because the two differ, d_{eff} is a descriptive rescaling of the window exponent, not a property of the stimulus set. The nearest-neighbour dimension, which is a property of the stimuli, is also below the nominal d (Section 4.1), and this is consistent with codes below the border exceeding the nominal bound.

4.4 Gratings, $d = 1$

We give each number for ranks 5–30 and, in parentheses, for ranks 11–30. For every code of the family $c_k = (1 + k^2/\kappa^2)^{-(\nu+1/2)}$ that we computed, the 32-direction window exponent lies below α_∞ , by 0.402 to 4.446 (0.502 to 4.180) (Figure 2). The shortfall has two parts. The operator spectrum is itself shallower than α_∞ over the window, by 0.135 to 4.253 (0.086 to 3.750), because it is flat for frequencies below κ and because each frequency carries two equal eigenvalues; even an exact power law $c_k = k^{-3}$ gives 2.821 (2.881). Sampling at 32 directions folds the frequencies above 16 onto lower ones and lowers the exponent by a further 0.104 to 0.344 (0.288 to 0.636) (aliasing). The border codes give 0.228 to 2.598 (0.321 to 2.405), all below 3, while differentiable codes with $\nu = 1.5$ give 0.386 to 3.592 (0.558 to 3.468) and exceed 3 for $\kappa \leq 2$ in both windows. Within this family and over the bandwidths computed, a 32-direction exponent above 3, such as 3.43, comes only from differentiable codes, in either window. Outside the family it does not. In the example of Figure 2, two codes share the $\nu = 1.5$, $\kappa = 1$ spectrum up to frequency 1,000 and continue as n^{-2} (not

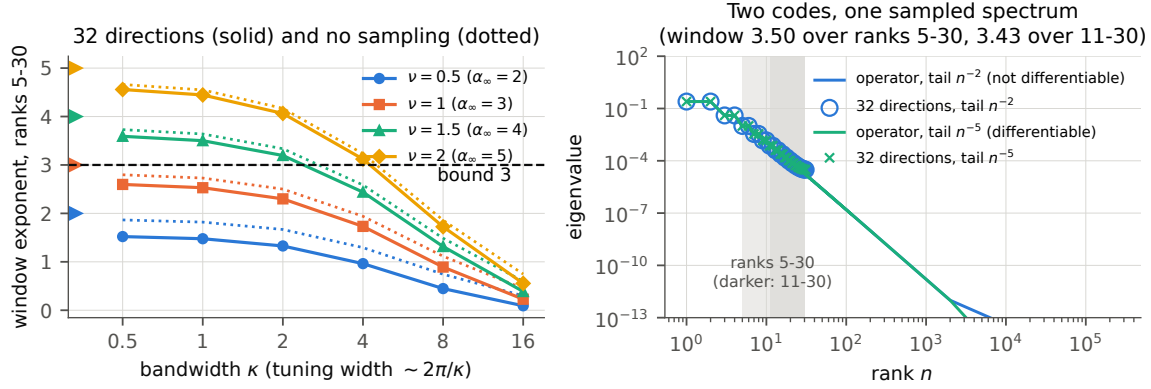


Figure 2: Left: ranks 5–30 window exponent (the window of the Methods of [1]; the text gives ranks 11–30, the window of their deposited code) for 32 equally spaced directions (solid) and for the operator spectrum without sampling (dotted), against the bandwidth κ of the tuning; arrowheads: $\alpha_\infty = 1 + 2\nu$; dashed: the bound 3. Right: the numerical example of Proposition 1. Two codes share the differentiable $\nu = 1.5$, $\kappa = 1$ spectrum up to frequency 1,000 and continue as n^{-2} (not differentiable) or n^{-5} (differentiable). Lines: operator spectra; markers: their 32-direction spectra, which coincide; both have window exponent 3.5012 over ranks 5–30 and 3.4306 over ranks 11–30, above the bound 3.

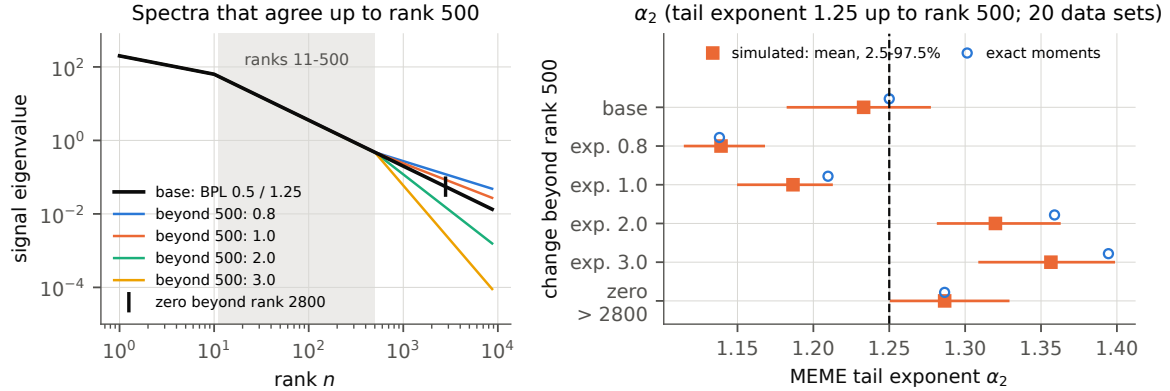


Figure 3: Left: the base spectrum, a broken power law with exponents 0.5 and 1.25, and variants that agree with it up to rank 500; shaded: ranks 11–500. Right: MEME α_2 (diagonal weights) for each spectrum; squares and bars, mean and 2.5th–97.5th percentiles over 20 simulated data sets with common random numbers; circles, the fit to the exact moments. Dashed: the base tail exponent 1.25.

differentiable, Section 2) or n^{-5} ; their tail masses are 2.0×10^{-9} and 5.0×10^{-10} , their 32-direction eigenvalues differ by at most 1.5×10^{-6} of the 30th eigenvalue, and both have window exponent 3.5012 (3.4306), the value of the family code: a non-differentiable code gives a 32-direction exponent above the bound in either window. Finite populations of 8,704 neurons changed the 32-direction exponent by at most 0.028 (0.044) on average, with a standard deviation of at most 0.007 (0.012).

4.5 The broken-power-law tail exponent

The variants in Table 4 agree with the base spectrum, a broken power law with tail exponent 1.25, up to rank 500. Beyond it they decay with exponent 0.8, 1.0, 2.0 or 3.0, or stop. Their exact-moment tail exponents range from 1.138 to 1.394 (from 1.405 to 1.552 for the base with 1.5): α_2 moves

Table 4: MEME broken-power-law tail exponent α_2 for spectra that differ from the base broken power law beyond rank 500 (a different exponent, or zero after rank 2,800) or at every rank (a floor adding 5% or 15% of the base variance, or a factor $e^{-n/1000}$). Window: ranks 11–500 exponent of the true spectrum. Trace: variance relative to the base. Exact: fit to the exact moments, diagonal weights (diag.) and full whitening (full). Simulated (diagonal weights): mean $\pm$ standard error over 20 data sets, and the paired difference from the base with its 95% interval. Table 5 gives how many of the data sets the misfit flags. Last column: the same variants of a base with $\alpha_2 = 1.5$, exact moments, diagonal weights.

change	window	trace	base $\alpha_2 = 1.25$				$\alpha_2 = 1.5$
			exact		simulated		exact
			diag.	full	α_2	paired difference	
none	1.250	1.000	1.250	1.250	1.233 $\pm$ 0.006		1.500
exponent 0.8 after 500	1.250	1.142	1.138	1.144	1.139 $\pm$ 0.004	−0.094 [−0.103, −0.085]	1.405
exponent 1.0 after 500	1.250	1.064	1.210	1.207	1.187 $\pm$ 0.004	−0.047 [−0.054, −0.039]	1.442
exponent 2.0 after 500	1.250	0.914	1.359	1.350	1.320 $\pm$ 0.006	+0.087 [+0.077, +0.097]	1.529
exponent 3.0 after 500	1.250	0.879	1.394	1.386	1.357 $\pm$ 0.007	+0.123 [+0.113, +0.134]	1.552
zero after 2,800	1.250	0.950	1.286	1.289	1.287 $\pm$ 0.006	+0.053 [+0.044, +0.063]	1.518
floor 5%	1.243	1.050	1.218	1.216			1.447
floor 15%	1.230	1.150	1.135	1.141			1.311
factor $e^{-n/1000}$	1.353	0.832	1.438	1.429			1.594

Table 5: Simulated data sets flagged by the eigenmoment misfit, out of 20 per spectrum, with the exact (Clopper–Pearson) 95% interval for the flag probability. A data set is flagged when its misfit exceeds the 95th percentile of a reference distribution of misfits of the correctly specified base. Diagonal weights, pooled: the 100 cross-fitted base data sets. Full whitening: the same pooled reference; the first four folds of it (base data sets 0–79); the first fold (the 20 paired base data sets); and leave one out (each of the 20 paired data sets whitened with the covariance of the other 19 paired base data sets; reference, the 20 base misfits computed the same way).

change beyond rank 500	diagonal weights		full whitening			
	pooled	pooled	first four folds	first fold	leave one out	
none	1 [0.001, 0.249]	0 [0.000, 0.168]	1 [0.001, 0.249]	1 [0.001, 0.249]	1 [0.001, 0.249]	
exponent 0.8 after 500	0 [0.000, 0.168]	0 [0.000, 0.168]	2 [0.012, 0.317]	4 [0.057, 0.437]	5 [0.087, 0.491]	
exponent 1.0 after 500	0 [0.000, 0.168]	0 [0.000, 0.168]	1 [0.001, 0.249]	2 [0.012, 0.317]	1 [0.001, 0.249]	
exponent 2.0 after 500	0 [0.000, 0.168]	0 [0.000, 0.168]	0 [0.000, 0.168]	2 [0.012, 0.317]	2 [0.012, 0.317]	
exponent 3.0 after 500	0 [0.000, 0.168]	0 [0.000, 0.168]	0 [0.000, 0.168]	3 [0.032, 0.379]	3 [0.032, 0.379]	
zero after 2,800	0 [0.000, 0.168]	0 [0.000, 0.168]	0 [0.000, 0.168]	3 [0.032, 0.379]	2 [0.012, 0.317]	

toward the true continuation, by at most 0.14, and stays far from it. No window of ranks up to 500 sees these changes. The eigenmoments see them through the trace, which changes by the factor 0.88 to 1.14; one data set registers this (the log trace has a standard deviation of 0.008), so the variance in the tail is observed. The moments $p \geq 2$ change by at most 0.01 standard deviations. The trace fixes how much variance lies in the tail, not how it decays. With diagonal weights the broken power law absorbs the change into α_2 : the exact-moment misfits are at most 0.009, against a null median of 0.007 and 95th percentile of 0.033 over 100 simulated data sets of the correctly specified model, and no data set of any variant is flagged (of the base, 1 of 20); the median misfits of the variants, 0.004 to 0.007, are even below that of the base, 0.010. With full-covariance whitening, closer to the check that Pospisil and Pillow propose (Section 1), the misfit of the correctly specified model has a heavy upper tail: median 3.80 and 95th percentile 16.51 over the 100 base data sets, against

3.36 and 9.49 for a chi-square distribution with 4 degrees of freedom (eight moments, four fitted parameters counting the profiled break). The five folds have medians 3.30, 3.19, 3.76, 3.25, 8.39; the last contributes 4 of the 5 misfits above the pooled 95th percentile, and the largest misfit is 136. The far-tail variants raise the median misfit of the simulated data sets from 3.30 to at most 5.52. How many of their data sets are flagged depends on the reference (Table 5): none of 20 against the pooled 95th percentile, 0 to 2 of 20 against the 95th percentile of the first four folds (10.20), 2 to 4 against that of the first fold, the 20 paired base data sets (8.13), and 1 to 5 with the covariance estimated from only 19 data sets (the other paired sets, leaving one out), the exponent-0.8 variant most often; under the last two references the base itself is flagged in one of 20. With diagonal weights, and with full whitening against the pooled reference, no variant data set is flagged, which excludes a flag probability above 0.168 for each variant (95% Clopper–Pearson bound). Under the other references the largest count, 5 of 20, has the 95% interval 0.087 to 0.491, so there a flag probability up to 0.491 is not excluded. What these 20 data sets show, then, is that the misfit detected the far-tail changes in at most 5 of 20 cases under any reference, and in none with diagonal weights. The whitened fits shift α_2 by similar amounts (exact moments: +0.136 for exponent 3.0, −0.106 for 0.8).

In the simulated data (Figure 3) α_2 of the base is 1.233 ± 0.006 , with a standard deviation over data sets of 0.026. Against the exact-moment value 1.250 this is a difference of −0.017, or −2.9 standard errors; over all 100 base data sets, each fitted with weights from the other four fifths, α_2 is 1.240 ± 0.002 , a difference of −0.010, or −4.2 standard errors. With diagonal weights our implementation is therefore biased low in this simulator. With full whitening we detect no bias: 1.254 ± 0.004 (1.0 standard errors) on the paired data sets and 1.251 ± 0.002 (0.5) on all 100, against 1.250. The bias is smaller than every shift below, and a bias common to the base and a variant cancels in their paired difference. The far-tail variants shift it by −0.094 to +0.123 (paired differences). The largest shift, +0.123 [+0.113, +0.134] for exponent 3.0, is 4.8 times that standard deviation and 0.62 of the gap of 0.20 between Pospisil and Pillow’s average α_2 of 1.20 and the natural-image bound of 1; for exponent 0.8 the shift is −0.094 [−0.103, −0.085]. Common random numbers helped only moderately: across data sets the α_2 of a variant correlates with that of the base at 0.66 to 0.76, and the paired differences have standard deviations of 0.017 to 0.022 against 0.018 to 0.031 for α_2 itself. Changes that also touch ranks below 500 can move α_2 further: a floor adding 15% of the base variance lowers the window exponent only to 1.230 but gives $\alpha_2 = 1.135$ (1.311 around 1.5).

For the Matérn codes of Section 2 the relevant input is the set of eigenmoment estimates MEME would compute from their noise-free responses at the 2,800 stimuli (Section 3); these estimate the population moments, not those of the 2,800-stimulus sample spectrum. The weights and the misfit threshold come from the simulated broken power law with neural noise, so for these noise-free moments the misfit is a heuristic filter, not a calibrated test. Many of these spectra are not broken power laws, and the diagonal-weight misfit shows it: its median over the 90 codes computed is 0.08, far above the null. For $\ell \geq 1$ the profiled break sits at the lower edge of its grid, rank 4, in 59 of 60 fits with either weighting, and at $\ell = 1/4$ it sits at the upper edge, rank 30, in 3 diagonal-weight fits and 1 full-whitening fit; one diagonal-weight fit ends at the upper limit $\alpha_2 = 6$. These α_2 are single-exponent fits from rank 4 onward rather than tail exponents in the usual sense. Among the 30 fits whose diagonal-weight misfit lies within the null 95th percentile, 7 of the 12 differentiable codes ($\nu = 1.5$) have α_2 below the bound and 2 of the 6 codes with $\nu = 0.75$ have α_2 above it, and the border codes fall below it in 6 and above it in 6 cases. The full-whitening misfit passes a different set of 62 fits: 6 of its 24 differentiable codes lie below the bound, 6 of its 17 codes with $\nu = 0.75$ above it, and its border codes 6 below and 15 above. Under either check, the tail exponents of passing fits lie on both sides of the bound for every smoothness computed. These moments are for infinitely many neurons, while the fit runs over 8,704 indices; for a population of N Gaussian-process

neurons $\mathbb{E} \text{tr} \Sigma^2$ exceeds the infinite-population value by a relative amount of about PR/N , with $\text{PR} = (\text{tr} \Sigma)^2 / \text{tr} \Sigma^2$ the participation ratio. PR reaches 864 at $\ell = 1/4$ in the 8D sets and is at most 38 for $\ell \geq 1$ (a relative change of at most 0.4%). With the moments of a population of 8,704 neurons at $\ell = 1/4$, $\log \text{tr} \Sigma^2$ rises by up to 0.105 (2.6 weight standard deviations) and α_2 changes by at most 0.058; every one of these fits stays below the bound, the number that pass rises from 17 to 23 of 30 with diagonal weights and from 18 to 28 with full whitening, and the statement about both sides of the bound still holds. These fits cover a restricted grid ($\nu \in \{0.75, 1, 1.5\}$, $\ell \in \{1/4, 1, 4\}$).

4.6 cvPCA bias and the noise level

For the simulator with the broken power law $\alpha_2 = 1.5$ (realized window slope 1.497) the bias of the cvPCA window exponent was -0.123 ± 0.003 with the calibrated noise, $+0.020 \pm 0.001$ with the non-aligned noise divided by 4, and $+0.007 \pm 0.001$ with it divided by 16. None of the three conditions matches the calibration recording: its cvPCA spectrum (Section 3) at ranks 11, 100 and 500 was (1.9, 7.3, 7.9), (1.7, 7.0, 10.9) and (1.8, 5.9, 13.3) times the simulated one in the three conditions, so in every condition the simulated cvPCA spectrum lies below the recording’s at these ranks. The bias shrinks in magnitude as the noise is reduced, but we computed no signal-to-noise ratio of the tail, for the simulations or for any recording, so these biases cannot be transferred to the data. It does not bear on identifiability, which concerns the noise-free spectrum.

5 Discussion

What the reported exponents can show. The exponents 1.49, 1.65 and 3.43 of [1] are descriptive statistics of particular stimulus sets over particular rank windows. They are consistent with the bound, since codes that satisfy it can produce them. They cannot show that the code satisfies it, nor the stronger conclusions that it is “as high-dimensional as possible” or that the asymptotic exponent is close to $1 + 2/d$ (the fitted exponents are “near but above the lower bounds”; the objection is to reading them as the asymptotic exponent), because codes that violate the bound, or sit exactly at it, produce the same window exponents. Border codes with 8,704 neurons reach 1.49 in three of the six 8D sets in the unwhitened coordinates (in whitened ones none does with infinitely many neurons) and 1.65 in all four 4D sets; codes below the border exceed the bound in some settings (Section 4); and on the torus a continuously differentiable code and one with infinite expected squared gradient can have finite-sample spectra that differ by less than any given amount at every rank, for every stimulus set (Corollary 1). Conversely, a window exponent below $1 + 2/d$ would not show that a code is non-differentiable, since smooth codes with short length scales give such values. The ordinal prediction, larger exponents for lower d , is reproduced by codes of fixed smoothness, so its confirmation does not bear on smoothness either. A window exponent measures how the code’s variance is spread over the modes the stimulus set resolves, at scales set by the tuning. This bears on the reanalysis of Davidovich and Roudi [5], who found the 8D exponents above the bound “with a reasonable margin” and wrote that the uncertainty of the fits “does not affect the conclusions regarding the smoothness of the population response to low-dimensional stimuli”: their result concerns the fitted exponent, and our computations say that even an exactly known window exponent does not decide smoothness. The broken-power-law tail exponent is likewise set by the assumed continuation beyond the resolved ranks: a tail that departs from the assumed form changes the trace, the fit absorbs this into α_2 , and in our simulations the misfit flagged at most 5 of 20 data sets of any variant: none with diagonal weights (a flag probability above 0.168 excluded at 95%), and with full whitening from none to 5 depending on the reference distribution and the covariance estimate (95% upper bounds up to 0.491). Pospisil and Pillow show that MEME recovers

eigenvalues beyond the rank of the data when the parametric form is right (their Supplementary Fig. S1, “Demonstration that MEME estimator can infer eigenvalues beyond the rank of data”); they also note that the form of the tail “remains unclear”.

What a test would need. One route is many more stimuli per dimension, with the window moved to higher ranks as well: as P grows, the fixed ranks 11–500 converge to the operator eigenvalues at those ranks, which for long tuning length scales remain pre-asymptotic, so more stimuli alone do not bring the window into the asymptotic regime. How many would be needed depends on the unknown length scale. A heuristic shows the scale of the problem: the nearest-neighbour spacing of P stimuli in d dimensions shrinks as $P^{-1/d}$, so resolving scales ten times shorter than 2,800 stimuli resolve now takes of the order of 10^d times as many stimuli. At $d = 1$ the obstacle is the number of directions: with 32 the window ends at the sampling frequency. The other route is a parametric model of the code, for example a kernel family such as (2) fitted to how response similarity falls with stimulus distance, with the bound read off the fitted smoothness. Such a test holds within the model, depends on the metric in which the code is assumed stationary, and meets the same resolution limit, because smoothness is decided at the shortest distances the stimulus set contains. Either route needs its assumptions stated in advance, and the geometry of each stimulus set, not only the nominal d , must enter the analysis.

Limitations. The codes placed on the stimulus coordinates are isotropic, stationary Gaussian-process codes in the Euclidean metric of the image principal components, or of their whitened version. Real codes are neither isotropic nor stationary, and the metric in which the neural code is smooth is unknown; the family shows what the window exponent can do at fixed smoothness, not how visual cortex behaves. Proposition 1 is proved for bounded eigenfunctions, and for the Matérn codes on $\mathbb{R}^d$ we rely on computation. The identifiability computations are noise-free; estimation error comes on top. The MEME computations use one implementation with stated deviations from the authors’ code, one noise model with fixed neurons and illustrative spectra, and for the Matérn codes weights and a misfit threshold calibrated on a different spectrum; the size of the shifts in α_2 depends on these choices, their existence does not. The stimulus-count dependence rests on 4 subsets per size. We make no statement about the exponents of the 8D, 4D or grating recordings. The bound of [1] is a theorem, and nothing here questions it; the question is only whether data of this size can show that a code satisfies it.

Data and code availability. The stimulus image files are part of the deposit of Stringer et al. on figshare, <https://doi.org/10.25378/janelia.6845348> (license CC BY-NC 4.0). The note’s source folder lists the image files used, with their checksums, and holds the Python programs that produce every number, table and figure, their outputs and the commands to rerun them; it has not yet been deposited in a public archive.

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References

- [1] C. Stringer, M. Pachitariu, N. Steinmetz, M. Carandini and K. D. Harris, High-dimensional geometry of population responses in visual cortex, *Nature* **571** (2019) 361–365; doi:10.1038/s41586-019-1346-5. Author manuscript PMC6642054; Supplementary Information read in part (Thms. 1, 2, 4, 5; §§2.1, 2.7, 3.5).
- [2] C. Stringer, M. Pachitariu, M. Carandini and K. Harris, Responses of ten thousand neurons to 2,800 natural images, figshare (2018); doi:10.25378/janelia.6845348.v3 (reference [36] of [1]).

- [3] MouseLand, `stringer-pachitariu-et-al-2018b`, github.com/MouseLand/stringer-pachitariu-et-al-2018b, commit 58443d1 (read: `python/utils.py`, `processResp/compileResps.m`, `powerlaws/get_powerlaw.m`, and the fitting windows in `mainfigs/fig3.m` and `powerlaws/statsShuffledPCA.m`).
- [4] D. A. Pospisil and J. W. Pillow, Revisiting the high-dimensional geometry of population responses in the visual cortex, *Proc. Natl. Acad. Sci. USA* **122** (2025) e2506535122; doi:10.1073/pnas.2506535122. Correction: e2533074122. SI Appendix read in part (captions of Figs. S1, S2).
- [5] I. A. Davidovich and Y. Roudi, Bayesian interpolation for power laws in neural data analysis, preprint, arXiv:2204.08525 (2022).
- [6] V. Koltchinskii and E. Giné, Random matrix approximation of spectra of integral operators, *Bernoulli* **6** (2000) 113–167 (abstract read).
- [7] M. L. Braun, Accurate error bounds for the eigenvalues of the kernel matrix, *J. Mach. Learn. Res.* **7** (2006) 2303–2328.
- [8] J. Shawe-Taylor, C. K. I. Williams, N. Cristianini and J. Kandola, On the eigenspectrum of the Gram matrix and the generalization error of kernel-PCA, *IEEE Trans. Inf. Theory* **51** (2005) 2510–2522 (read: Sects. I–III).
- [9] W. Kong and G. Valiant, Spectrum estimation from samples, *Ann. Statist.* **45** (2017) 2218–2247 (read: arXiv:1602.00061v5, Sects. 1 and 3, Algorithm 1).
- [10] S. Spigler, M. Geiger and M. Wyart, Asymptotic learning curves of kernel methods: empirical data versus teacher–student paradigm, preprint, arXiv:1905.10843v8 (2020) (read: Sects. 1 and 7).
- [11] D. A. Pospisil, `revisit_hid_geom`, github.com/dapospisil/revisit_hid_geom, commit afcd301 (read: `src/eig_mom.py`).